

Portfolio Comparative Analysis Report

12-ETF Institutional Portfolios vs 3-ETF Test Portfolio (SPY / TLT / GLD)

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Portfolio Definitions

12-ETF Group (Institutional Model Portfolios)

Three portfolios derived from the Vise Target Allocation ETF model (2025-05-20):

Portfolio	Description	# Tickers
40/60 (TA ETF)	40% equity / 60% bond allocation	18
100/0 (TA ETF)	100% equity allocation	9
2-ETF (IVV/EFA)	Concentrated stress-test (US equity + Intl equity)	

3-ETF Test Portfolio

A deliberately cross-asset portfolio designed to test comparative analysis behavior across uncorrelated exposures:

Ticker	Weight	Asset Class
SPY	40%	US Broad Equity (S&P 500)
TLT	40%	US Long-Term Treasury Bonds (20+ yr)
GLD	20%	Gold — inflation hedge / crisis diversifier

Simulation Parameters

Parameter	Value
Initial Capital	\$1,000,000
Market Periods	Bear (2007–2008), Baseline (2010–2019), Bull (2023–2024)
Strategies	Monthly, Quarterly, Yearly rebal + Abs/Rel threshold bands
TLH Threshold	10% loss
Tax Rates	ST: 35% / LT: 20%
Transaction Costs	12 bps (5 commission + 5 slippage + 2 bid-ask)
Total Simulation Runs	192 (144 for 12-ETF group, 48 for 3-ETF)

Section 1 — Overall Performance (All Strategies, All Periods)

Metric	12-ETF	3-ETF	Advantage
Avg CAGR	0.97%	7.77%	3-ETF +6.80%
Avg Sharpe Ratio	0.42	0.93	3-ETF +0.51
Avg Max Drawdown	-24.05%	-12.91%	3-ETF (shallower by 11%)
Avg Volatility (Ann)	16.58%	9.37%	3-ETF (lower by 7.2%)
Avg Total Return	20.65%	29.02%	3-ETF +8.4%
Avg Tracking Error	0.75%	0.71%	3-ETF (marginally tighter)
Avg Information Ratio	-0.95	+0.19	3-ETF (positive vs negative)

Key takeaway: On a simple average across all periods and strategies, the 3-ETF portfolio outperforms on every risk-adjusted metric. However, this aggregate view is heavily influenced by the Bear Market period — the picture changes significantly when broken out by regime.

Section 2 — Performance by Market Regime

Bear Market (2007–2008)

Metric	12-ETF	3-ETF
Avg CAGR	-19.95%	+2.88%
Avg Sharpe Ratio	-0.79	+0.25
Avg Max Drawdown	-44.8%	-19.4%
Avg Volatility	26.9%	11.7%
Best Strategy (Sharpe)	Quarterly Rebal + TLH (−0.62)	Rel Threshold 50% + TLH (+0.41)

Verdict: 3-ETF wins decisively. TLT and GLD both surged during the 2007–2008 financial crisis. The 12-ETF portfolios held international equity and emerging market exposure which were all hit simultaneously — no true diversification. The 3-ETF's 44.8 ppt max drawdown advantage is the single largest gap in the entire analysis.

Baseline Market (2010–2019)

Metric	12-ETF	3-ETF
Avg CAGR	7.04%	9.51%
Avg Sharpe Ratio	0.65	1.41
Avg Max Drawdown	-17.4%	-8.2%

Metric	12-ETF	3-ETF
Avg Volatility	11.8%	6.8%
Best Strategy (Sharpe)	Yearly Rebal + TLH (1.06)	Quarterly Rebal + TLH ^{**} (1.90) ^{**}

Verdict: 3-ETF wins. The decade-long post-crisis recovery favored equities, but SPY's growth combined with TLT's steady income and GLD's stability produced a smoother, higher Sharpe path. The 12-ETF portfolios had more dispersion across many correlated equity sleeves.

Bull Market (2023–2024)

Metric	12-ETF	3-ETF
Avg CAGR	15.83%	10.94%
Avg Sharpe Ratio	1.40	1.13
Avg Max Drawdown	-9.95%	-11.16%
Avg Volatility	10.96%	9.66%
Best Strategy (Sharpe)	Abs Threshold 10% / No TLH ^{**} (1.73) ^{**}	Abs Threshold 20% + TLH (1.27)

Verdict: 12-ETF wins. This is the only regime where the 12-ETF outperforms. In a strong equity bull market, TLT and GLD become dead weight — TLT underperformed significantly in the 2023–2024 rate environment. The 12-ETF's equity concentration (especially the 100/0 portfolio) captures full upside. The 5% CAGR gap here is meaningful.

Section 3 — Tax-Loss Harvesting Analysis

All figures based on TLH-enabled runs only.

Core TLH Metrics

Metric	12-ETF	3-ETF	Winner
Avg Tax Paid	+\$5,213	-\$844	3-ETF
Avg Losses Harvested	\$77,654	\$46,452	12-ETF (raw volume)
Avg TLH Event Count	10.96	2.38	12-ETF (more events)
Avg Execution Costs	\$3,649	\$2,346	3-ETF
Avg Tax Alpha 2	\$740	\$9,624	3-ETF (13x higher)
Avg Loss Carryforward ST	\$3,345	\$15,486	3-ETF

Harvest Efficiency

The 12-ETF harvests more dollars but converts them to tax value less effectively:

Portfolio	Losses Harvested	Tax Paid	Efficiency Ratio
12-ETF	\$77,654	+\$5,213	14.9x
3-ETF	\$46,452	-\$844	55.0x

The 3-ETF's net tax paid is **negative** — harvested losses fully offset all realized gains and carry \$15,486 forward as future tax shields. The 12-ETF still writes a check to the IRS despite harvesting nearly twice as much.

CAGR Lift from TLH (TLH On minus TLH Off)

Market Period	12-ETF Lift	3-ETF Lift
Bear Market	-1.07%	+1.00%
Baseline Market	-0.38%	+4.62%
Bull Market	-0.59%	+0.40%
Overall Average	-0.68%	+2.00%

Critical finding: TLH hurts performance in the 12-ETF portfolios across every single market period. For the 3-ETF, TLH adds value in every period — most dramatically in the Baseline decade (+4.62% CAGR lift).

Why TLH Works Better in the 3-ETF

- **Fewer events, higher quality.** With only 2–3 TLH harvests per run vs 11 for the 12-ETF, each harvest is more impactful and generates less cost drag.
- **No wash-sale interference.** SPY, TLT, and GLD are in different asset classes — harvesting one has no risk of triggering a wash sale on the others. The 12-ETF's many equity tickers (IVV, IVW, IVE, OEF, EFA, EFV, EFG) are highly correlated and wash sales limit harvest effectiveness.
- **Lower execution costs.** 3-ETF: \$2,346 vs 12-ETF: \$3,649 average per run.

Top TLH Runs by Tax Alpha 2

12-ETF Group:

Portfolio	Period	Strategy	Tax Alpha 2	Losses Harvested	TLH Events
2-ETF (IVV/EFA)	Bear Market	Yearly Rebal + TLH	\$17,449	\$66,004	2
2-ETF (IVV/EFA)	Bear Market	Abs Threshold 10% + TLH	\$17,269	\$65,076	1
2-ETF (IVV/EFA)	Bear Market	Abs Threshold 20% + TLH	\$17,269	\$65,076	1

3-ETF Portfolio:

Portfolio	Period	Strategy	Tax Alpha 2	Losses Harvested	TLH Events
3-ETF (SPY/TLT/GLD)	Bear Market	Quarterly Rebal + TLH	\$15,744	\$47,984	5
3-ETF (SPY/TLT/GLD)	Bear Market	Abs Threshold 5% + TLH	\$15,075	\$58,490	3

Portfolio	Period	Strategy	Tax Alpha 2	Losses Harvested	TLH Events
3-ETF (SPY/TLT/GLD)	Bear Market	Rel Threshold 25% + TLH	\$14,380	\$55,954	3

Note: The best individual TLH runs in the 12-ETF group come from the concentrated 2-ETF (IVV/EFA) sub-portfolio, not the diversified 40/60 or 100/0 models — further evidence that fewer, larger positions produce better TLH outcomes.

Section 4 — Best Strategy by Period

Period	Portfolio	Best Strategy	Sharpe	CAGR
Bear Market	3-ETF	Rel Threshold 50% + TLH	0.41	4.71%
Bear Market	12-ETF	Quarterly Rebal + TLH	-0.62	-19.81%
Baseline Market	3-ETF	Quarterly Rebal + TLH	1.90	12.33%
Baseline Market	12-ETF	Yearly Rebal + TLH	1.06	6.66%
Bull Market	12-ETF	Abs Threshold 10% / No TLH	1.73	22.13%
Bull Market	3-ETF	Abs Threshold 20% + TLH	1.27	11.96%

Section 5 — Summary Scorecard

Category	Winner	Margin
Crisis protection (Bear Market)	3-ETF	Large
Long-run risk-adjusted return (Baseline)	3-ETF	Large
Bull market return capture	12-ETF	Moderate
Volatility (lower is better)	3-ETF	Large
Max drawdown (shallower is better)	3-ETF	Large
Raw losses harvested	12-ETF	Moderate
TLH efficiency (Tax Alpha 2)	3-ETF	Very Large (13x)
TLH CAGR impact	3-ETF	3-ETF +2.00% vs 12-ETF -0.68%
Net tax outcome	3-ETF	Negative tax paid vs +\$5,213
Execution costs	3-ETF	Moderate

Final Score: 3-ETF wins 9 of 10 categories. The 12-ETF only wins on raw losses harvested and bull-market return capture.

Section 6 — Key Takeaways

- **The 3-ETF is a superior TLH vehicle.** Despite harvesting fewer raw dollars, it converts losses into real after-tax value at 55x efficiency vs 14.9x for the 12-ETF. TLH adds +2% CAGR to the 3-ETF; it subtracts -0.68% from the 12-ETF.
- **Portfolio concentration improves TLH quality.** The 12-ETF's best TLH results come from its most concentrated sub-portfolio (2-ETF: IVV/EFA), not from the broader institutional models. More tickers = more wash-sale risk and cost drag.
- **Asset class diversity matters more than ticker diversity.** The 3-ETF holds 3 tickers but 3 genuinely uncorrelated asset classes. The 12-ETF holds many tickers but they are largely correlated equity and bond sub-factors.
- **Crisis regime defines the long-run winner.** The 3-ETF's Bear Market performance (+2.88% CAGR vs -19.95%) dominates the aggregate comparison. Protecting capital in a crash compounds forward far more powerfully than marginal bull-market outperformance.
- **The 12-ETF is better suited for investors with pure equity bull-market conviction.** Its 22% CAGR in the best Bull Market run beats the 3-ETF's 12% — but that advantage reverses dramatically in any crisis.
- **TLH in the 12-ETF needs redesign.** The consistent negative CAGR drag from TLH across all periods suggests the current thresholds and wash-sale handling are sub-optimal at scale. Fewer, higher-conviction harvests would likely perform better.

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